BENZATHINE BENZYLPENICILLIN

Deep IM only — never IV

Lasts 2–4 weeks

CHEMICAL NAME N,N′-dibenzylethylenediamine salt of benzylpenicillin (long-acting depot penicillin G)

CLASS Long-acting depot penicillin (penicillin G benzathine)

TRADE NAMES Bicillin L-A · Extencilline · Retarpen · Penadur

THERAPEUTIC USES / INDICATIONS

- Syphilis (primary, secondary, early latent; late latent with 3 doses).
- Secondary prevention of rheumatic fever (every 3–4 weeks).
- Streptococcal pharyngitis (single dose) when oral compliance uncertain.

PHARMACODYNAMICS

- Benzylpenicillin inhibits **cell-wall synthesis** → bactericidal against *Treponema pallidum* and streptococci.

ADMINISTRATION

- Deep IM only into upper outer gluteal or anterolateral thigh (children).
- Never IV — cardiac arrest and death reported.
- Aspirate before injecting; inject slowly; shake suspension well.
- May mix with lidocaine diluent (where licensed) to reduce pain.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Pain and induration at injection site.
- Fever, headache.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis (prolonged exposure — observe 30 min).
- Accidental IV/intra-arterial injection: cardiac arrest, gangrene, nerve damage (Hoigne's/Nicolau syndrome).
- Jarisch–Herxheimer reaction in syphilis.

DOSAGES

- **Early syphilis:** 2.4 million units IM single dose (1.2 MU in each buttock).
- **Late latent syphilis:** 2.4 MU weekly for 3 weeks.
- **Rheumatic fever prophylaxis:** 1.2 MU every 3–4 weeks (children <30 kg: 600,000 units).
- **Strep pharyngitis:** 1.2 MU once (600,000 units if <27 kg).

CONTRAINDICATIONS & CAUTIONS

- Penicillin hypersensitivity.
- IV administration.

KEY INTERACTIONS

- **Probenecid:** ↑ levels.
- **Tetracyclines:** possible antagonism.

PHARMACOKINETICS

- Slowly released from IM depot → low but prolonged blood levels.
- Detectable levels for 2–4 weeks.
- Excreted in urine.

NURSING MANAGEMENT

- Check allergy history; have adrenaline ready; observe 30 min after injection.
- Use large-bore needle; aspirate; inject slowly and deep.
- Record date of next dose (rheumatic fever schedule).
- Syphilis: contact tracing and follow-up serology.
- Monitor site for pain or blanching.

PATIENT EDUCATION

- Keep every appointment for repeat injections.
- Injection site may be sore for a day or two.
- Fever after the first syphilis dose is expected.
- Partners need testing and treatment.

KEY POINTS

- Long-acting depot penicillin — deep IM only.
- Syphilis 2.4 MU; rheumatic fever every 3–4 weeks.
- Never IV; fatal.
- Observe 30 min for anaphylaxis.

SOURCES CHECKED **[115]** Benzylpenicillin benzathine 1.2 Million I.U. powder for suspension for injection (UK SmPC) **[116]** Bicillin L-A, IM (US label) *Full references p. 432–457*